

Assignment 2

CST - 109

Design and implement a simple cloud-based distributed file system. Implement the following:

1. Basic Distributed File System (Without Replication)

Implement a client, a metadata server, and multiple storage servers.

The client should be able to upload and download files.

2. Implement file replication

Each file chunk must be replicated on at least 2 or 3 different storage servers.

Metadata server must maintain the primary and replica locations for each chunk

Clients must be able to download a file from any available replica.

3. Fault Tolerance and Node failure handling

Simulate failures by randomly shutting down storage servers and evaluate system behavior.

Randomly stop 1–2 storage servers during runtime.

Clients should still be able to download files using replicas.

Metadata server should detect failed nodes.

Share the log files showing file uploads and downloads, chunk distribution across servers, replication details, requests served by each storage server and also share screenshots of running components and failure scenarios as supporting evidence.